

$$\sum_{n=1}^{\infty} \frac{(x+3)^n}{n^2}$$

$$\lim_{n \rightarrow \infty} \left| \frac{(x+3)^{n+1}}{(n+1)^2} \cdot \frac{n^2}{(x+3)^n} \right| = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^2 \cdot |x+3|$$

$$= |x+3| < 1 \Rightarrow R=1$$

$x = -2$ $\sum_{n=1}^{\infty} \frac{1}{n^2} \rightarrow$ yakınsaktır $\rightarrow p=2 > 1$ old.

$$-4 < x < -2$$

$$-4 \leq x \leq -2$$

$$[-4, -2]$$

$x = -4$ $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \rightarrow$ mutlak yakınsaktır

$$\sum_{n=1}^{\infty} \frac{(2n-1)^{2n} (x-1)^n}{(3n-2)^{2n}}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{(2n-1)^{2n} (x-1)^n}{(3n-2)^{2n}} \right|} = \lim_{n \rightarrow \infty} \left(\frac{2n-1}{3n-2} \right)^2 \cdot |x-1|$$

$$\sqrt[n]{|x-1|^n} = \sqrt[n]{|x-1|^n} = |x-1|$$

$$= \frac{4}{9} \cdot |x-1| < 1$$

$x = \frac{13}{4}$ in $\sum_{n=1}^{\infty} \frac{(2n-1)^{2n} \cdot \left(\frac{9}{4}\right)^n}{(3n-2)^{2n}} \rightarrow \left(\frac{3}{2}\right)^{2n}$

$$= \sum_{n=1}^{\infty} \left(\frac{6n-3}{6n-4} \right)^{2n}$$

$$|x-1| < \frac{9}{4} \rightarrow R = \frac{9}{4}$$

$$-\frac{5}{4} < x < \frac{13}{4}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{6n-3}{6n-4} \right)^{2n}} = \lim_{n \rightarrow \infty} \left(\frac{6n-3}{6n-4} \right)^2 = 1 \not< 1 \rightarrow \text{ratio test}$$

$$-\frac{5}{4} < x < \frac{13}{4}$$

$x = -\frac{5}{4}$ in $\sum_{n=1}^{\infty} (-1)^n \cdot \left(\frac{6n-3}{6n-4} \right)^2$

$a_{n+1} > a_n \rightarrow \text{ratio test}$

$$f(x) = \frac{3}{(1-x)(1+2x)} = \frac{A}{1-x} + \frac{B}{1+2x}$$

$$3 = \underbrace{(A+B)}_3 + \underbrace{(2A-B)}_0 x$$

$$A+B=3 \quad B=2A$$

$$A=1 \\ B=2$$

$$f(x) = \frac{1}{1-x} + 2 \cdot \frac{1}{1+2x}$$

$$= \sum_{n=0}^{\infty} x^n + 2 \cdot \sum_{n=0}^{\infty} (-1)^n (2x)^n = \sum_{n=0}^{\infty} [x^n + 2 \cdot (-1)^n \cdot 2^n \cdot x^n]$$

$$= \sum_{n=0}^{\infty} (1 + (-1)^n \cdot 2^{n+1}) \cdot x^n$$

$$f(x) = \sin 3x + x \cos 3x$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

①

$$\sum_{n=0}^{\infty} \frac{(-1)^n \cdot (3x)^{2n+1}}{(2n+1)!} = \sin 3x = 3x - \frac{(3x)^3}{3!} + \frac{(3x)^5}{5!} - \frac{(3x)^7}{7!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\cos 3x = 1 - \frac{(3x)^2}{2!} + \frac{(3x)^4}{4!} - \frac{(3x)^6}{6!} + \dots = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{(3x)^{2n}}{(2n)!}$$

$$x \cdot \cos 3x = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{3^{2n} \cdot x^{2n+1}}{(2n)!} \quad \text{②}$$

$$\sum_{n=0}^{\infty} (-1)^n \cdot 3^{2n} \cdot x^{2n+1} \cdot \left[\frac{3}{(2n+1)!} + \frac{1}{(2n)!} \right]$$

$$f(x) = xe^{-2x}$$

$$f(x) = e^{x^2}$$

$$\rightarrow e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$e^{-2x} = \sum_{n=0}^{\infty} \frac{(-2x)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-2)^n \cdot x^n}{n!}$$

$$f(x) = 2x$$

$$f(x^2) = 2x^2$$

$$x \cdot e^{-2x} = \sum_{n=0}^{\infty} \frac{(-2)^n \cdot x^{n+1}}{n!}$$

$$e^{x^2} = \sum_{n=0}^{\infty} \frac{(x^2)^n}{n!} = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$$

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} = \lim_{n \rightarrow \infty} \frac{\cancel{n}}{\cancel{n} \cdot \underbrace{\left(1 + \frac{1}{n}\right)}} = 1$$